

Project ID:

RP-026-079

1. Topic (12 words max)

A Smart IOT-Driven Digital Platform for Sustainable Dry Fish Processing in Sri Lanka

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	RP-026-079
Year	2026

5. Brief description of the research problem, including references (300 words max) – references not included in word count.

The dry fish industry is an important component of Sri Lanka's coastal economy and food security, providing an affordable protein source and supporting many small-scale fishers and processors. Yet, dry fish processing is largely informal and manually managed. Drying is commonly carried out in open or semi-open environments, exposing fish to dust, insects, animals, smoke, and unstable weather, including sudden rainfall that can re-wet partially dried fish and increase spoilage risk. This results in inconsistent quality, contamination risks, and measurable post-harvest losses. Although controlled drying approaches (e.g., solar-dryer based methods) are discussed as a way to reduce exposure-related quality issues compared to open drying [1], most small-scale processors still rely on manual. Food safety concerns in fish products may include risks related to histamine and other biogenic amines when handling and processing are poor [4], reducing consumer confidence and market value. In addition, meeting microbiological limits and hygiene expectations is difficult in fully manual settings.

These challenges have multidimensional impacts. Socially, workers (often women) face occupational health issues due to long exposure to heat, sun and smoke, as well as unhygienic working environments. Economically, producers experience unstable income due to spoilage, weather dependency, limited access to buyers, and price exploitation by intermediaries. Environmentally, unmanaged fish waste and inefficient energy practices contribute to coastal pollution and local emissions.

Technically, the industry lacks real-time monitoring, data-driven decision support, and an integrated digital platform that connects drying condition tracking with quality assurance and traceability. IoT-based monitoring and control of drying systems have been proposed in recent work to improve drying process visibility and consistency [2], while digital technologies are increasingly highlighted as important for improving transparency and

compliance in fish supply chains [3]. This research proposes a locally contextualized, software centric, IoT-enabled digital platform to improve drying quality control, reduce losses, enhance fair market access, and support sustainability and traceability, aligning with SDG 2 (Zero-Hunger), SDG 9 (Industry, Innovation and Infrastructure), SDG 12 (Responsible Consumption and Production), and SDG 14 (Life Below Water).

References:

- [1] Food and Agriculture Organization of the United Nations (FAO), “Improved fish drying using polyethylene solar dryer,” FAO Fisheries & Aquaculture (OpenASFA record), n.d. [Online]. Available: <https://www.fao.org/fishery/ar/openasfa/96817841-af8b-4ce8-acfc-2bec83b8906b>.
- [2] N. Mishra, S. K. Jain, N. Agrawal, N. K. Jain, N. Wadhawan, and N. L. Panwar, “Development of drying system by using internet of things for food quality monitoring and controlling,” *Energy Nexus*, vol. 11, Art. no. 100219, Sep. 2023, doi: 10.1016/j.nexus.2023.100219. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2772427123000499>.
- [3] J. Cromwell, C. Turkson, M. Dora, and F. A. Yamoah, “Digital technologies for traceability and transparency in the global fish supply chains: A systematic review and future directions,” *Marine Policy*, vol. 178, Art. no. 106700, Aug. 2025, doi: 10.1016/j.marpol.2025.106700. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0308597X25001150>.
- [4] World Health Organization (WHO) and Food and Agriculture Organization of the United Nations (FAO), *Joint FAO/WHO Expert Meeting on the Public Health Risks of Histamine and Other Biogenic Amines from Fish and Fishery Products: Meeting Report*. Geneva, Switzerland: WHO, 2013. [Online]. Available: <https://www.who.int/publications/i/item/9789240691919>.

6. Brief description of Existing Research and Systems (300 words max)

Existing research on dried fish and food drying consistently highlights that controlled drying environments (such as enclosed dryers and solar dryers) can improve hygiene and final product quality compared to traditional open drying. These approaches reduce direct exposure to dust, insects, animals, and rain, and help lower contamination and post-harvest losses [1]. Food safety literature also emphasizes that poor handling and improper processing conditions in fish products can lead to chemical and microbiological hazards, reinforcing the need for better control and monitoring during processing [4].

From a technology perspective, many studies have explored IoT-based drying systems where sensors track microclimate conditions such as temperature and humidity, enabling remote monitoring and alerts. Research shows that IoT-enabled monitoring and control can improve drying consistency and provide early warning for unsafe conditions [2]. In parallel, predictive analytics and machine learning have been used in food processing to estimate drying completion time and detect patterns linked to spoilage risk, supporting more proactive decisions compared to manual checks.

Separately, digital systems for seafood supply chains increasingly use tools such as QR codes, digital records, and in some cases blockchain-based approaches to improve traceability and transparency. Reviews note that traceability technologies are important for building trust, supporting compliance, and improving accountability across supply chains [3].

However, most existing solutions are fragmented; some focus only on dryer hardware and sensors, while others address traceability or market information separately. In Sri Lanka, solutions are often centered on physical drying improvements, while software-driven integration across monitoring, analytics, sustainability, and traceability remains limited.

This highlights a research gap for a locally feasible, end-to-end digital platform that combines real-time monitoring, predictive decision support, and traceability features under practical constraints such as cost sensitivity and limited connectivity.

References:

- [1] Food and Agriculture Organization of the United Nations (FAO), “Improved fish drying using polyethylene solar dryer,” FAO Fisheries & Aquaculture (OpenASFA record), n.d. [Online]. Available: <https://www.fao.org/fishery/ar/openasfa/96817841-af8b-4ce8-acfc-2bec83b8906b>.
- [2] N. Mishra, S. K. Jain, N. Agrawal, N. K. Jain, N. Wadhawan, and N. L. Panwar, “Development of drying system by using internet of things for food quality monitoring and controlling,” *Energy Nexus*, vol. 11, Art. no. 100219, Sep. 2023, doi: 10.1016/j.nexus.2023.100219. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2772427123000499>.
- [3] J. Cromwell, C. Turkson, M. Dora, and F. A. Yamoah, “Digital technologies for traceability and transparency in the global fish supply chains: A systematic review and future directions,” *Marine Policy*, vol. 178, Art. no. 106700, Aug. 2025, doi: 10.1016/j.marpol.2025.106700. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0308597X25001150>.
- [4] World Health Organization (WHO) and Food and Agriculture Organization of the United Nations (FAO), *Joint FAO/WHO Expert Meeting on the Public Health Risks of Histamine and Other Biogenic Amines from Fish and Fishery Products: Meeting Report*. Geneva, Switzerland: WHO, 2013. [Online]. Available: <https://www.who.int/publications/i/item/9789240691919>.

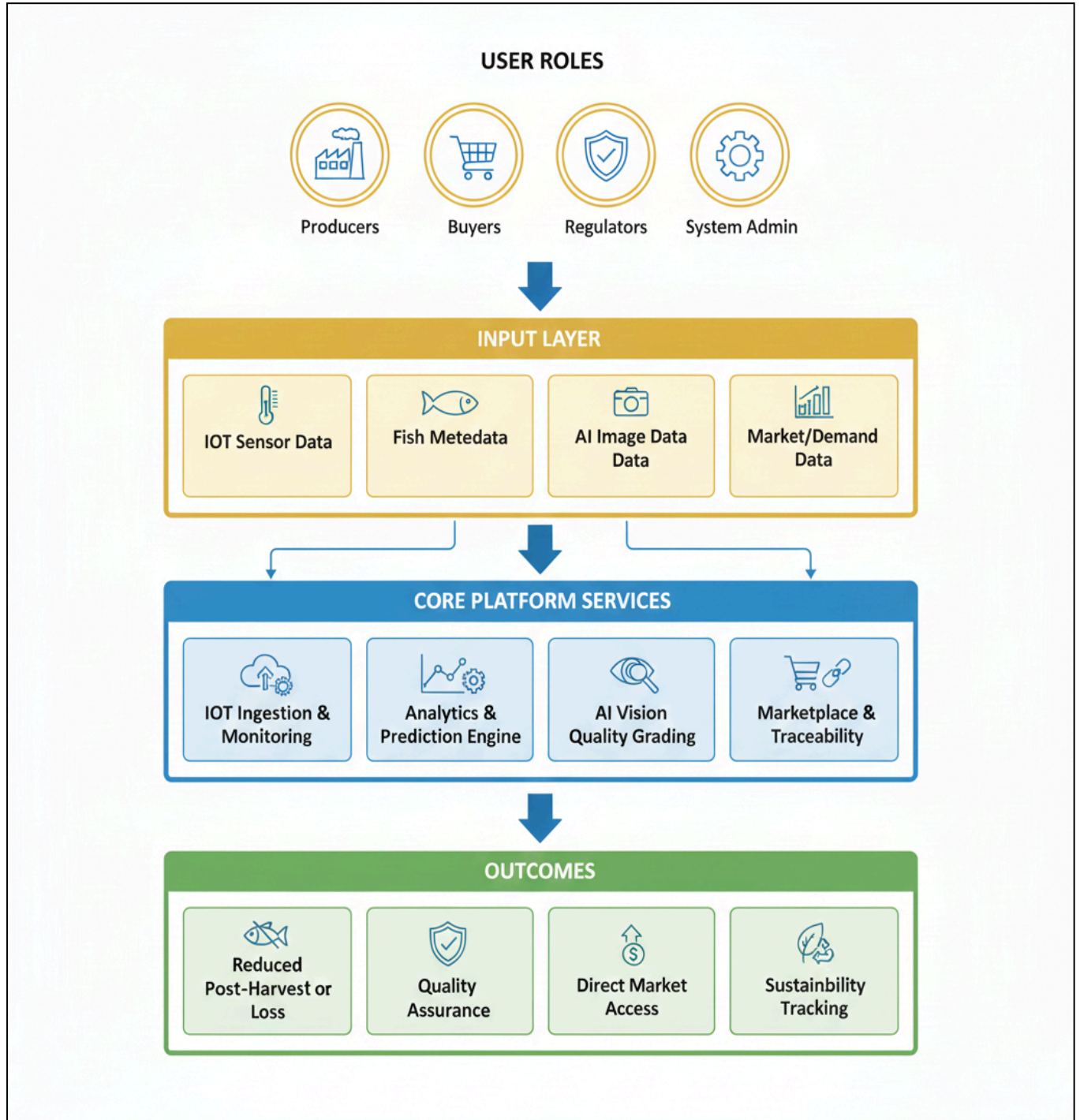
7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution is a smart IoT-driven platform for fry fish processing in Sri Lanka. It digitizes and optimizes the dry fish life cycle by integrating sensor-based monitoring, predictive analytics, market connectivity, sustainability tracking and traceability into a single software platform.

System Perspective (IPO model)

The system collects IoT sensor data including temperature, humidity, drying duration, and optionally smoke levels, air quality, and batch weight, which feeds into real-time microclimate monitoring with threshold-based alerts to deliver improved quality consistency and reduced contamination risk across the drying process. Fish batch metadata covering fish type, batch size, and drying stage undergoes data analytics to forecast drying completion times and assess spoilage risk, ultimately reducing post-harvest losses and ensuring optimal processing conditions. Producer and site information is integrated with a digital marketplace module that connects producers directly with buyers to improve price transparency, resulting in improved producer income through fairer market access and better negotiating power. Market price and buyer demand data powers the marketplace platform's matching algorithms and pricing recommendations, leading to reduced post-harvest losses and improved producer income via more efficient and equitable market connections. Waste generation and handling data is processed through waste tracking algorithms that provide recommendations for safer handling and potential reuse opportunities, generating sustainability insights that help minimize environmental impact. Finally, user consent and traceability records are transformed through secure traceability record creation systems that assign batch IDs and track processing milestones with controlled access, producing a transparent supply chain that supports regulatory compliance and builds trust among all stakeholders.

The system is designed with module services (IoT ingestion, analytics, marketplace, traceability) to allow phased adoption and future integration with regulators and certification bodies



8. Brief overview of the availability of necessary specialised domain expertise, knowledge, and data. (500 words max)

This research is supported by the availability of relevant technical expertise, accessible domain knowledge, and practical data collection pathways suitable for building and evaluating a prototype system.

IoT Microclimate Monitoring and Quality Alerts

Expertise & Knowledge: The required knowledge for IoT prototyping and software development is widely established in software engineering and embedded systems. Low-cost devices (e.g., ESP32-class microcontrollers) and standard sensors (temperature and humidity) are well documented and commonly used in environmental monitoring applications.

Data Availability: Primary data is obtainable through a small-scale drying setup that logs temperature, humidity, and drying duration over time. Optional additions such as batch weight and smoke/air-quality can be included depending on device availability. These logs support evaluation of alert accuracy and sensor uptime.

Predictive Analytics for Drying Completion and Spoilage Risk

Expertise & Knowledge: Applied analytics for time-series prediction is well supported by mature machine learning toolkits and published research on feature engineering and forecasting. Standard evaluation approaches (e.g., MAE for time prediction and F1-score for risk classification) are widely used and appropriate for prototype-level validation.

Data Availability: A practical pilot dataset can be created from controlled drying trials (e.g., 30–60 batches) with time-series sensor logs and labeled outcomes (completion time, acceptable vs. spoiled/defective). Where real-world data is limited, controlled simulations and realistic synthetic datasets can be used to test algorithm performance and stability.

Computer Vision–Based Quality Grading and Contamination Detection

Expertise & Knowledge: Methods for image classification and defect detection using convolutional neural networks (CNNs) are well established in applied AI research, and suitable training and evaluation practices are widely available.

Data Availability: Images of dried fish batches can be collected using mobile phones and labeled into quality grades and defect categories (e.g., discoloration, mold-like appearance, uneven drying, insect contamination signs). A feasible pilot target is approximately 300 - 500 labeled images, including samples captured under different lighting conditions.

Sustainability Tracking, Traceability, and Ethical Data Handling

Expertise & Knowledge: Traceability systems (batch IDs, milestone logs) and secure access control are standard software engineering practices. Domain knowledge on hygienic seafood processing, controlled drying methods, and Sri Lankan food safety expectations is accessible through published literature and public guidance.

Data Availability: Waste generation and handling practices can be captured through short surveys and interviews with processors in coastal communities. Publicly available regulatory and guidance documents support compliance mapping. Ethical requirements can be addressed through informed consent, minimal data collection, secure storage, and role-based access control within the platform.

9. Objectives and Novelty

Main Objective To design and develop a smart, software-centric, IoT-enabled digital platform that improves dry fish quality control, post-harvest loss reduction, sustainability practices, and end-to-end traceability within Sri Lanka's dry fish processing industry. The platform integrates real-time sensor monitoring, AI/ML analytics, computer vision-based grading, and traceability + sustainability records into a single system that can be adopted in phases and used in low-resource, coastal processing settings.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
VIVIPEM L B R V IT22639844	To predict drying completion time and spoilage risk using sensor time-series data, reducing post-harvest losses by enabling proactive, evidence-based decisions during drying.	Time-series dataset creation: - Collect continuous sensor logs (temperature, humidity, time; optional: batch weight/air quality if available). - Record outcomes for each batch: completion time, acceptable quality vs. defective/spoiled. Feature engineering and signal extraction: - Convert raw sensor streams into meaningful indicators (e.g., rolling averages, humidity trend slopes, time above risk threshold, day/night variations, sudden humidity jump events).	This introduces predictive analytics to an industry that currently relies on guesswork and manual inspection. The innovation is not only collecting sensor data, but transforming it into actionable predictions (when drying will finish and whether spoilage risk is rising). It supports proactive decisions under Sri Lanka's variable coastal weather , making the drying process more stable, reducing waste, and improving income reliability for small producers.

		- Link these features to batch outcomes. Predictive model development: - Train machine learning models for: Completion prediction (regression): estimate remaining drying time. Spoilage risk (classification): low/medium/high or safe/risky based on patterns. - Use model selection appropriate for small datasets (e.g., Random Forest / Gradient Boosting) with careful validation. Evaluation and impact measurement: - Evaluate completion prediction using MAE (hours error). - Evaluate spoilage classification using F1-score, precision, and recall. - Demonstrate practical value: earlier intervention	
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		(cover/extend/relocate batches) and reduced wastage. Platform integration: • Show predicted completion time and risk score on the dashboard alongside live sensor data and alerts. • Store predictions as part of the batch record for traceability and analysis.	
GUNASEKARA D T C D P IT22127396	To develop a computer vision module that automatically grades dry fish quality and detects visible contamination/defects to reduce subjective manual grading and strengthen quality assurance.	Image dataset collection and labelling: • Capture batch images under real conditions (different lighting, angles, backgrounds). • Define labelling categories such as: ○ Quality grades (A/B/C) or acceptable/reject. ○ Defect labels: discolouration, uneven drying, mould-like growth, and insect contamination signs.	This introduces AI-based automated visual grading to Sri Lankan dry fish processing, where inspection is currently subjective and varies from person to person. The module provides repeatable, data-driven grading and early detection of visible defects that often lead to rejected products and consumer distrust. Recording grading results into traceability logs also enables quality verification and supports future compliance needs.

		Define grading criteria (repeatable rules): • Create clear defect criteria and grading standards that match practical processing expectations. • Ensure the labels are consistent and usable for training. CNN model training and evaluation: • Train a Convolutional Neural Network (CNN) for classification/detection tasks. • Evaluate using accuracy and F1-score, plus a confusion matrix to see which defects are misclassified. • Test robustness under varying lighting conditions, because real-world drying sites are not controlled environments. Workflow integration in the platform: • Integrate a “capture/upload image”	
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		function into the platform. • Display grading results on the batch dashboard (grade + detected defects). • Store results inside the batch traceability record as evidence of quality checks.	
PESHALA G A K J IT22554536	To implement sustainability and traceability mechanisms that support responsible processing by tracking waste events and maintaining secure, end-to-end batch records.	Waste event modeling (sustainability tracking): • Define waste categories (e.g., fish offcuts, spoiled batches, wastewater/saline runoff, packaging waste). • Create a structured method to record waste events: type, quantity estimate, time, disposal method, and location. Recommendations for handling/reuse: • Provide system-guided suggestions such as safer disposal, reuse options,	This component integrates sustainability tracking + traceability into an informal supply chain where records are usually missing or paper-based. It transforms the process into a transparent, auditable workflow, supporting safer waste practices and enabling trustworthy batch histories. Even without physical waste sensors, structured digital waste logs and traceability provide practical sustainability insights and strengthen the platform's readiness for future certification or regulatory reporting.

		or basic best practices (where applicable and realistic). ● Maintain a sustainability dashboard summarising waste generation patterns. Traceability record design (end-to-end batch history): ● Create unique Batch IDs and a traceability timeline (catch/receipt → drying stage updates → monitoring logs → AI grading result → outcome). ● Ensure traceability records store key milestones and evidence, not just text. Consent and access control: ● Implement role-based permissions (e.g., processor/admin/regulator-future). ● Store consent and ensure data access is controlled and logged to protect users and build trust. Usability evaluation: ● Assess whether users can easily create batch records, log waste events, and view batch	
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		histories without confusion.	
SANJAYA E W M IT22189394	To design and evaluate a smart drying condition monitoring module that supports quality assurance by continuously tracking drying microclimate conditions and providing early warnings when conditions become unsafe or unstable.	Prototype device setup (IoT layer): - Build a low-cost monitoring device using an ESP32-class microcontroller and sensors (minimum: temperature + humidity). - Deploy it near the drying area (or inside a prototype dryer) to continuously capture drying environment conditions at fixed intervals. Threshold and rule definition (quality control logic): - Define safe/unsafe ranges for temperature and relative humidity based on drying requirements and hygiene considerations. - Create rule sets such as: “humidity above X for Y minutes” or “rapid humidity rise indicating rain exposure.”	This component introduces microclimate-driven, real-time quality monitoring into a sector where drying is traditionally managed by experience and visual judgement. Instead of reacting after spoilage occurs, it supports early detection of high-risk drying conditions (e.g., humidity spikes from rainfall, prolonged high moisture). This creates a practical bridge between traditional processing and data-driven quality assurance , directly addressing inconsistency and contamination risks in manual drying operations.

		Real-time dashboard and alerts: • Develop a live dashboard displaying current readings, trends, and batch status. • Implement threshold-based notifications (web/mobile) for events such as high humidity, sudden changes, sensor downtime, or risky drying periods. Validation and usefulness evaluation: • Evaluate alert precision (how often alerts are correct) and recall (how often risky events are detected). • Collect feedback from users or test operators on whether alerts improve decisions (e.g., covering fish, extending drying, improving storage).	
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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

- a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes ☒ No ☐

- b) The proposed topic exhibits novelty.

Yes ☒ No ☐

- c) The student group can successfully execute the proposed project.

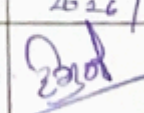
Yes ☒ No ☐

- d) The proposed sub-objectives reflect the students' areas of specialization.

Yes ☒ No ☐

- e) Any other comments:

Accepted with minor changes

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Jenny	Krishara	 2026/01/13
Co-Supervisor	Dr.	Dinuka	Wijendra	 2026/01/14
External Supervisor	Mr	Dinuth	Jayasekara	

Summary of external supervisor's (if any) experience and expertise
 The external supervisor owns a dry fish business and has practical experience in fish drying, quality control, spoilage prevention, and supply chain operation